

01 GET /alarms

RULES ["compatibility aliases derive only from canonical fields", "source R03 requires alarm_type OOS, rule_code R03_CONSEC, and value null", "source SUMMARY requires rule_code SUMMARY_OOC", "source TRACE requires rule_code TRACE_OOS"]

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {
      "area": {
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        "required": false,
        "schema": {
          "enum": ["ALL", "Etch", "Photo"],
          "type": "string"
        },
        "chamber": {
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          "required": false,
          "schema": {
            "type": "string"
          },
          "date_from": {
            "nullable": true,
            "required": false,
            "schema": {
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              "type": "string"
            },
            "date_to": {
            "nullable": true,
            "required": false,
            "schema": {
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              "type": "string"
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            "nullable": true,
            "required": false,
            "schema": {
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            },
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            "nullable": false,
            "required": false,
            "schema": {
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              "type": "boolean"
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            "parameter": {
            "nullable": true,
            "required": false,
            "schema": {
              "type": "string"
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            "source": {
            "nullable": true,
            "required": true,
            "schema": {
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              "type": "string"
            },
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                    "fields": {
                      "action_code": {
                        "nullable": true,
                        "required": false,
                        "schema": {
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                          "type": "string"
                        },
                      "alarm_id": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "alarm_type": {
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                          "type": "string"
                        },
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                          "type": "string"
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                          "type": "string"
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                        "nullable": true,
                        "required": false,
                        "schema": {
                          "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"],
                          "type": "string"
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                        "nullable": true,
                        "required": false,
                        "schema": {
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "schema": {
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                          "type": "string"
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                        "nullable": false,
                        "required": true,
                        "schema": {
                          "type": "string"
                        },
                      "mes_status": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"],
                          "type": "string"
                        },
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                        "required": true,
                        "schema": {
                          "type": "boolean"
                        },
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                        "required": false,
                        "schema": {
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                        },
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "nullable": false,
                        "required": true,
                        "schema": {
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                          "type": "string"
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                          "type": "string"
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                        },
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                      "recipe_step_no": {
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                        "schema": {
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                          "type": "integer"
                        },
                      "rule_code": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "enum": ["R03_CONSEC", "SUMMARY_OOC", "TRACE_OOS"],
                          "type": "string"
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                        "required": false,
                        "schema": {
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                          "type": "integer"
                        },
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                        "required": true,
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                        "required": false,
                        "schema": {
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                        "schema": {
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                        },
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                        "schema": {
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                                            },
                                            {
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                }
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    }
  }
}
```

02 GET /trace

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {
      "chamber": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "lot": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "parameter": {
        "nullable": false,
        "required": true,
        "schema": {
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          "type": "string"
        },
      "wafer": {
        "nullable": false,
        "required": true,
        "schema": {
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          "type": "string"
        }
      }
    },
    "responses": {
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            "fields": {
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                "nullable": false,
                "required": true,
                "schema": {
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                  "type": "string"
                },
              "recipe_step_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 1.0,
                  "type": "integer"
                },
              "seq_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 0.0,
                  "type": "integer"
                },
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                "nullable": false,
                "required": true,
                "schema": {
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                      "required": false,
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                          "fields": {
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                              "required": true,
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                                      "type": "integer"
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                                    },
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                                  "type": "array",
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                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

03 GET /parameters

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item
```

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s": {"additional_properties": false, "fields": {"LCL": {"nullable": true, "required": false, "schema": {"type": "number"}}, "LSL": {"nullable": true, "required": false, "schema": {"type": "number"}}, "TARGET": {"nullable": false, "required": true, "schema": {"type": "number"}}, "UCL": {"nullable": false, "required": true, "schema": {"type": "number"}}, "USL": {"nullable": false, "required": true, "schema": {"type": "number"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "ctrl_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "ctrl_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "spec_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "spec_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "target_value": {"nullable": false, "required": true, "schema": {"type": "number"}}, "unit": {"nullable": true, "required": false, "schema": {"type": "string"}}, "upper_only": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object", "type": "array", "shape": "array"}}
```

04 POST /documents/search

RULES []

```
{request": {"body": {"additional_properties": false, "fields": {"doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "query": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "top_k": {"nullable": false, "required": false, "schema": {"default": 4, "maximum": 10.0, "minimum": 1.0, "type": "integer"}}, "type": "object"}, "header": {"path": {"query": {"responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "content": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "score": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": -1.0, "type": "number"}}, "section": {"nullable": true, "required": false, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "array", "shape": "array"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

05 GET /relations/chambers/{chamber_id}

RULES [{"node_count equals len(nodes)", "relationship_count equals len(relationships)"]

```
{request": {"body": null, "header": {"path": {"chamber_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "query": {"include_production_context": {"nullable": false, "required": false, "schema": {"default": false, "type": "boolean"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "display_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "type": "object"}, "type": "array"}}, "production_context": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"returned_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "truncated": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "target": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "root_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

06 GET /agent/runs

RULES []

```
{request": {"body": null, "header": {"path": {"query": {"date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "predicted_fault_code": {"nullable": true, "required": false, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"action_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_id"
```

```
: {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_source": {
  "nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}},
"approval_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {
  "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {
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```

07 POST /agent/ask

```
RULES [{"fault_code and ground_truth_fault_code are forbidden", "metrology.alarm_result is forbidden"}]
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```

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```

08 GET /approvals

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields not null", "only EQP_HOLD actions have approvals"]

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item s": {
            "additional_properties": false,
            "fields": {
              "action_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "EQP_HOLD",
                    "MONITORING",
                    "WARNING"
                  ],
                  "type": "string"
                },
                "action_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
                "agent_run_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
                "approval_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
                "approved_at": {
                "nullable": true,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
                "approved_by": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
                "chamber": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
                "chamber_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
                "created_at": {
                "nullable": false,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
                "decided_at": {
                "nullable": true,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
                "decided_by": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
                "decision_comment": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
                "equipment": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
                "equipment_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
                "fault_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "FOC",
                    "MFD",
                    "OTH",
                    "RFM",
                    "TMD"
                  ],
                  "type": "string"
                },
                "lot": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
                "lot_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
                "predicted_fault_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "FOC",
                    "MFD",
                    "OTH",
                    "RFM",
                    "TMD"
                  ],
                  "type": "string"
                },
                "reason": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
                "status": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "APPROVED",
                    "PENDING",
                    "REJECTED"
                  ],
                  "type": "string"
                },
                "type": "object",
                "array": true,
                "shape": "array"
              }
            }
          }
        }
      }
    }
  }
}
```

09 POST /approvals/{approval_id}/decision

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields not null", "only EQP_HOLD actions have approvals"]

```
{
  "request": {
    "body": {
      "additional_properties": false,
      "fields": {
        "decided_by": {
          "nullable": false,
          "required": true,
          "schema": {
            "max_length": 40,
            "min_length": 1,
            "type": "string"
          }
        },
        "decision": {
          "nullable": false,
          "required": true,
          "schema": {
            "enum": ["APPROVED", "REJECTED"],
            "type": "string"
          }
        },
        "decision_comment": {
          "nullable": true,
          "required": false,
          "schema": {
            "max_length": 1000,
            "type": "string"
          }
        },
        "object": {
          "header": {},
          "path": {
            "approval_id": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "string"
              }
            },
            "query": {},
            "responses": {
              "200": {
                "schema": {
                  "additional_properties": false,
                  "fields": {
                    "action_code": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "enum": ["EQP_HOLD", "MONITORING", "WARNING"],
                        "type": "string"
                      }
                    },
                    "action_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "agent_run_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "approval_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "approved_at": {
                      "nullable": true,
                      "required": true,
                      "schema": {
                        "format": "date-time",
                        "type": "string"
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

```
ring"}}, "approved_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

10 GET /audit-logs

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"actor_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "entity_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "entity_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "event_type": {"nullable": true, "required": false, "schema": {"type": "string"}}}, "responses": {"200": {"schema": {"items": {"additional_properties": true, "fields": {"actor": {"nullable": false, "required": true, "schema": {"type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}, "type": "array"}, "shape": "array"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

11 GET /dataset/bounds

RULES []

deferred inventory

12 GET /dashboard/summary

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"area": {"nullable": true, "required": false, "schema": {"enum": ["ALL", "Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "equipment_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "area": {"nullable": true, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "daily_trend": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"date": {"nullable": false, "required": true, "schema": {"format": "date", "type": "string"}}, "has_r03_consec": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "ooc_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "type": "array"}, "date_range": {"nullable": false, "required": true, "schema": {"items": {"format": "date", "type": "string"}, "type": "array"}}, "equipment_counts": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}
```

```
{}}, "hierarchy": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "ooc_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "pending_approvals": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, 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13 GET /alarms/{source}/{alarm_id}

RULES []

```
{ "request": { "body": null, "header": {}, "path": { "alarm_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "query": {}, "responses": { "200": { "schema": {"additional_properties": false, "fields": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "agent_run_status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": false, "schema": {"enum": ["APPROVED", "AUTO", "EXPIRED", "PENDING", "REJECTED"], "type": "string"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "detail": {"nullable": false, "required": true, "schema": {"type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "hit_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "incident": {"nul
```

```
label": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}}, "type": "object"}}, "judgement": {"nullable": false, "required": true, "schema": {"enum": ["OOC", "OOS"], "type": "string"}}, "latest_agent_run_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "lot_hist_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "recipe_step_name": {"nullable": true, "required": false, "schema": {"type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "rule_id": {"nullable": false, "required": true, "schema": {"enum": ["R01_OOS", "R02_OOC", "R03_CONSEC"], "type": "string"}}, "sensor_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "wafer_no": {"nullable": true, "required": false, "schema": {"minimum": 1.0, "type": "integer"}}, "type": "object", "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object", "shape": "object"}}
```

14 GET /alarms/paged

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": { "area": { "nullable": true, "required": false, "schema": { "enum": ["ALL", "Etch", "Photo"], "type": "string" } }, "chamber_id": { "nullable": true, "required": false, "schema": { "type": "string" } }, "date": { "nullable": true, "required": false, "schema": { "format": "date", "type": "string" } }, "equipment_id": { "nullable": true, "required": false, "schema": { "type": "string" } }, "judgement": { "nullable": true, "required": false, "schema": { "enum": ["OOC", "OOS"], "type": "string" } }, "page": { "nullable": false, "required": false, "schema": { "default": 1, "minimum": 1, "type": "integer" } }, "sensor_id": { "nullable": true, "required": false, "schema": { "type": "string" } }, "size": { "nullable": false, "required": false, "schema": { "default": 20, "maximum": 100, "minimum": 1, "type": "integer" } } }, "responses": { "200": { "schema": { "additional_properties": false, "fields": { "items": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": false, "fields": { "action_code": { "nullable": true, "required": false, "schema": { "enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string" } }, "action_id": { "nullable": true, "required": false, "schema": { "min_length": 1, "type": "string" } }, "agent_run_status": { "nullable": true, "required": false, "schema": { "enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string" } }, "alarm_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "approval_status": { "nullable": true, "required": false, "schema": { "enum": ["APPROVED", "AUTO", "EXPIRED", "PENDING", "REJECTED"], "type": "string" } }, "area": { "nullable": false, "required": true, "schema": { "enum": ["Etch", "Photo"], "type": "string" } }, "chamber_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "detail": { "nullable": false, "required": true, "schema": { "type": "string" } }, "equipment_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "hit_cnt": { "nullable": true, "required": false, "schema": { "minimum": 0.0, "type": "integer" } }, "incident": { "nullable": false, "required": true, "schema": { "additional_properties": false, "fields": { "chamber_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "lot_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "lot_hist_id": { "nullable": true, "required": false, "schema": { "min_length": 1, "type": "string" } }, "lot_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "occurred_at": { "nullable": false, "required": true, "schema": { "format": "date-time", "type": "string" } }, "recipe_step_name": { "nullable": true, "required": false, "schema": { "type": "string" } }, "recipe_step_no": { "nullable": false, "required": true, "schema": { "minimum": 1.0, "type": "integer" } }, "rule_id": { "nullable": false, "required": true, "schema": { "enum": ["R01_OOS", "R02_OOC", "R03_CONSEC"], "type": "string" } }, "sensor_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "source": { "nullable": false, "required": true, "schema": { "enum": ["R03", "SUMMARY", "TRACE"], "type": "string" } }, "wafer_no": { "nullable": true, "required": false, "schema": { "minimum": 1.0, "type": "integer" } }, "type": "object", "type": "array" }, "page": { "nullable": false, "required": true, "schema": { "minimum": 1.0, "type": "integer" } }, "size": { "nullable": false, "required": true, "schema": { "maximum": 100.0, "minimum": 1.0, "type": "integer" } }, "total": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer" } }, "type": "object", "shape": "object", "422": { "schema": { "additional_properties": true, "fields": { "detail": { "nullable": false, "required": false, "schema": { "items": { "additional_properties": true, "fields": { "loc": { "nullable": false, "required": true, "schema": { "type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": "object"}, "type": "array"}}, "type": "object", "shape": "object" } }
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15 GET /traces/catalog

RULES []

```
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```

```
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```

16 POST /traces/search

RULES []

```
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17 GET /relations/equipment/{equipment_id}

RULES []

deferred inventory

18 GET /documents/{document_id}

RULES []

```
{ "request": { "body": null, "header": {}, "path": { "document_id": { "nullable": false, "required": true, "schema": {"type": "string"}}, "query": {}}, "responses": { "200": { "schema": { "additional_properties": false, "fields": { "chunks": { "nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": { "chunk_id": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}
```

```
{
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19 POST /agent/runs

RULES []

{"request": {"body": {"additional_properties": false, "fields": {"alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, {"type": "object"}}, {"header": {}, {"path": {}, {"query": {}}, {"responses": {"202": {"schema": {"additional_properties": false, "fields": {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"alarm": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, {"status": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"type": "object"}, {"shape": "object"}, {"422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, {"msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"type": {"nullable": false, "required": true, "schema": {"type": "string"}}}], "type": "object"}, {"type": "array"}]}, "type": "object"}, {"shape": "object"}]}}, {"type": "object"}]]}

20 GET /agent/runs/{run_id}

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22 POST /agent/runs/{run_id}/retry
RULES []
deferred inventory
23 GET /agent/runs/paged
RULES []
deferred inventory
24 GET /approvals/paged
RULES []
deferred inventory
25 GET /actions
RULES [{"delivery_policy": "b3d9Kafka/MES Mock", "action_code": "EQP_HOLD", "monitoring": true, "warning": true, "approval_status": "PENDING", "rejection": false, "reason": "No decisions, preflight, second batch, unknown"}]
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26 GET /actions/{action_id}
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\udc2b9\udc778\ubcb18\ub824\ub00b7\ud655\udc778 \uae30\ub85d \uc5c6\udc774 \udc774\uba54\udc77c \ubcb0f \udc790\ub3d9 Kafka/MES Mock\udc744 \uc5f0\ub3d9\ud55c\ub2e4; \udc2b9\udc778 \uc0c1\ud0dc\ub294 null\udc774\ub2e4"]
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27 POST /actions/{action_id}/deliveries/{channel}/retry
RULES []
deferred inventory
28 POST /internal/actions/{action_id}/delivery
RULES ["FAILED requires error_code non-null and permits provider_message_id null", "SENT requires provide
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29 POST /analytics/query
RULES []
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30 POST /analytics/graph-query
RULES []
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31 POST /analytics/validate
RULES []
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32 GET /analytics/history
RULES []
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33 GET /analytics/evaluations
RULES []
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34 GET /audit-logs/paged

RULES []

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[illegible]

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AgentAskResponse

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AgentEvaluationResponse

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AgentPredictionDetailItem

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AgentRunAccepted

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AlarmEvidence

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AlarmItem
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            "schema": {
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          "error_msg": {
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            "schema": {
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            "schema": {
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            "required": true,
            "schema": {
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              "type": "array"
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            "required": true,
            "schema": {
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            "required": true,
            "schema": {
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                          "fields": {},
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ApprovalDecisionRequest

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{ "additional_properties": false, "fields": {"decided_by": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decision": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "REJECTED"], "type": "string"}}, "decision_comment": {"nullable": true, "required": false, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}
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ApprovalItem

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{ "additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approved_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "approved_by": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "rules": ["PENDING requires all decision fields null", "APPROVED or REJECTED requires decided_by and decided_at", "only EQP_HOLD actions have approvals"], "type": "object"}
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AskDocumentEvidenceAlias

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{ "additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "type": "object"}
```

AskToolItem

```
{ "additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
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AuditLogItem

```
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AuditLogPageResponse

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```

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ChamberGraphResponse
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ChamberRelationResponse
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}
ChartType
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ChatToolCallItem
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CrossCheck
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DeliveryResult
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DocumentAskEvidence

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DocumentChunkItem

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DocumentDetailResponse

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DocumentEvidence

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DocumentHit

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DocumentType

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ErrorCode

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ErrorResponse

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DETECTION_COMPLETED / LOT_HIST / A
AGENT_RUN_STARTED / AGENT_RUN / C
HYPOTHESIS_GENERATED / AGENT_RUN / C
APPROVAL_REQUESTED / APPROVAL / C
APPROVAL_DECIDED / APPROVAL / C
ACTION_SENT / ACTION / C delivery service
ACTION_SEND_FAILED / ACTION / C delivery service
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AGENT_RUN_COMPLETED / AGENT_RUN / C
AGENT_RUN_FAILED / AGENT_RUN / C